

# SILAS M. FALDE

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## EDUCATION

### University of Michigan School of Engineering

*Junior B.E.S. in Data Science, GPA: 3.09/4.00*

**Ann Arbor, MI**

*August 2023 – December 2026*

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

- Michigan Data Science Team (MDST) – Member 2 years
- Heartbeat (A Cappella) – Member 1 year

### Western High School

*GPA: 3.963*

**Parma, MI**

*August 2019 – May 2023*

- Men's Varsity Soccer – Captain 1 year, Member 2 years
- Select Choir – Member 3 years

## WORK EXPERIENCE

### Ally Financial, Inc.

*Intern, Data Management*

**Detroit, MI**

*May 2025 – August 2025*

- Created data dictionaries and data lineage documentation leveraging Python, enhancing data governance and compliance.
- Added data quality checks to existing data pipelines using Python and SQL, improving data integrity and reliability.
- Developed a data availability dashboard in Power BI to monitor data refresh rates and availability, ensuring timely access to critical data.

### Michigan Soccer Referee Association

*Certified Grassroots Soccer Referee*

**Ann Arbor, MI**

*March 2022 – Present*

- Officiating on a referee team to create a safe, fair, and fun environment for players, coaches, and spectators.
- Applying knowledge of the rules of soccer to facilitate the sport.

## RESEARCH

### University of Michigan Department of Epidemiology

*Research Assistant – Computational Lead*

**Ann Arbor, MI**

*September 2025 – Present*

- **Leading** the technical development of a **Python** pipeline using a large language model to classify narratives of collective violence.
- Performing data cleaning, exploratory data analysis including **Natural Language Processing**, and descriptive statistical analysis on a large dataset using **Python**.
- Evaluating model performance using **performance metrics** including accuracy, precision, recall, and F1 score.

### University of Michigan Department of Epidemiology

*Research Assistant*

**Ann Arbor, MI**

*September 2024 – May 2025*

- Awarded 3rd place (top ~1%) in the CDC Youth Mental Health: Novel Variables Data Challenge.
- Developed **Python** pipelines using a large language model on a high performance computing cluster to extract trends from the narratives of victims of suicide.
- Cleaned, preprocessed, and **feature engineered** a confidential dataset of over 400,000 narratives.
- Applied **natural language processing** techniques to identify relevant cases and enhance the annotation process.
- Documented findings with reports and visualizations to communicate findings to both the research team and the academic community through publication.

## PROJECTS

### Predicting Survival in the ICU

*Project Developer*

**Ann Arbor, MI**

*January 2025 – March 2025*

- Developed logistic regression and RNN models to predict survival of patients in the ICU using a public dataset using **Scikit-learn** and **PyTorch**.
- Performed data cleaning, preprocessing, and feature engineering to prepare the data for model training.
- Fine-tuned and evaluated the models using **cross-validation** and **grid search**.

## SKILLS

### Programming

C++, Python, R, SQL, Java

### Python

PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, NLTK, spaCy

### Software

Git, VS Code, Microsoft Office Suite, Google Suite, Snowflake, Collibra, Alation, Manta